MILITARY SPECIFICATION

WARHEAD, 155MM, M549A1
BONDING ASSEMBLY

This Amendment forms a part of Military Specification MIL-W-50577C (AR), dated 20 June, 1979, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

2.2: Delete ASTM "E18" and substitute:

"E103 Rapid Indentation Hardness Testing of Metallic Materials".

PAGE 5

* 3.11: Delete in its entirety and substitute the following:

"3.11 Slow bend precracked charpy test. The nominal fracture toughness shall be 25 KSI in. minimum."

PAGE 8

Table 1, under Warhead Unpainted Metal Parts, delete: "Interior Surface Test....4.5.6" and substitute: "Precracked Charpy Test 1 100% 3.11 4.5.4.3/Appendix".

PAGE 12

4.4.2.3: Delete Critical defect 1 in its entirety. Add Special defect A as follows:

"SPECIAL A precracked charpy test 4.4.3.1.4 3.11 4.5.4.3/Appendix"

PAGE 13

4.4.2.3: Delete Major defect 122 in its entirety.

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PAGE 17

4.4.2.7: Major defect 101, delete "100%" and substitute: "0.40%"

PAGE 18

4.4.3.1.1: Delete, "The test shall be conducted...samples of paragraph 4.4.3.1.2."

* 4.4.3.1.4 Delete in its entirety and substitute the following:

"4.4.3.1.4 Slow bend precracked charpy test. Beginning with the first heat treat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated warhead from each heat treated lot shall be subjected to the precrack charpy test. The warhead selected shall be of the highest average hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Two specimens shall be tested, with accpetance based on the fracture toughness average. In addition, no individual fracture toughness shall be less than 23 KSI $\sqrt{in.}$ If one test is invalid, acceptance may be based on the fracture toughness of the valid sample, or another sample from the same warhead may be tested. If either the average or individual fracture toughness is below the minimum allowable, the heat treated lot represented shall be rejected, or subjected to retemper and retest, for all mechanical properties, if approved by the Government. After three consecutive heat treated lots have been accepted, the test shall be discontinued. The test shall be re-instituted, until three consecutive heat treated lots have been complied with the applicable requirements, if any of the following occur:

a. Steel if procured from a different producer, or from a substantially different process as determined by the Government.

b. Forgings are procured from a different producer.

c. A substantial change is made in the forging process or configuration, as determined by the Government.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process, as determined by the Government."

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PAGE 19

4.4.3.4: Delete in its entirety and substitute the following:

"4.4.3.4 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of this applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490".

Add 4.4.5 as follows:

"4.4.5 Steel quality. Steel supplied shall comply with MIL-S-50783, except that macroetch quality shall be per MIL-STD-1459. The quality of the steel as indicated by the macroetch results shall be equal to, or better than A5, B3 and C8 with defects D1, D3, D4, D5, D6, D7 or D8 unacceptable. Manufacturer shall provide a certification that the applicable MIL-S-50783, and macroetch requirements have been met."

PAGE 20

4.5.3.2: Delete "Calibration" and substitute "Verification".

PAGE 21

4.5.3.2: Add the following to the end of the first paragraph:

"The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (See 4.5.3.4)."

Delete paragraph 4.5.3.4 and substitute the following in its place:

"4.5.3.4 Operator qualifications. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasoncis equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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4.5.4: Delete in its entirety and substitute the following:

"4.5.4 Mechanical properties.

4.5.4.1 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining to a sufficient depth, 0.015 inches minimum, to remove scale and get below the decarburization zone. The preparation process to be used shall require government approval. Manual grinding shall not be used. The surface preparation requirement may be eliminated if the following conditions are met:

1. Lab results confirm that current production process produces a decarburized zone of 0.003 inches maximum in the test area. Contractor shall provide samples, or photo-micrographs, of high and low hardness samples to the government for approval. Samples may be the same as for 4.5.4.2 below.

2. A controlled atmosphere is maintained in the austenitizing furnace.

3. Depth of decarubrization shall be reverified whenever precrack charpy testing is required (see 4.4.3.1.4), but only for the first heat treat lot.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

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d. The Hardness Tester shall use a standard 10MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

e. One reading shall be taken at location T and one at location X as indicated on Dwg. 9235995.

4.5.4.2 Tension. The two warheads selected for testing shall have two specimens taken from each and tested as follows:

a. Base section. The base section tensile specimen shall be taken transversely, one inch from the centerline. Failure to meet the yield strength or elongation requirements of dwg. 9235995 shall require a retest from the remaining portion of the base. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the heat treated inspection lot. In addition, failure of either individual test specimen to meet 130,000 psi minimum yield strength and 3% minimum elongation shall be cause for rejection of that heat treatment inspection lot represented by the sample.

b. Side wall. One specimen shall be taken from between stations T and X. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot.

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Testing shall be conducted in accordance with American Society for Testing and Materials, Method E8 with the largest substandard size round specimen possible.

4.5.4.3 Slow bend precrack charpy. The two specimens shall be taken from locations C1 and C2 on Figure 1. The test shall be conducted at a specimen temperature of -65 degree fahrenheit (^{O}F). The test shall be conducted in accordance with the procedure defined in the Appendix of this specification with the notches machined across the rear face of the specimen. Locations C3 and C4 on Figure 1 shall be used for replacement specimens for locations C1 and C2 respectively."

PAGE 22

4.5.8: line 6, delete "M185 or".

PAGE 23

4.5.9.1: Delete "Calibration" and substitute "Verification".

Add the following: "The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (See 4.5.9.3)."

PAGE 24

4.5.9.3: Delete in its entirety and substitute the following:

"4.5.9.3 Operator qualifications. For operation of eddy current equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of eddy current equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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4.5.10.1: Delete in its entirety and substitute the following:

"4.5.10.1 Salt spray testing. Salt spray testing shall be accomlished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.4 and test samples shall be dried in accordance with Drawing 9235995."

4.5.11: Delete in its entirety.

PAGE 25

* 6.3: Delete "ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801" and substitute "ARDEC, ATTN: AMSMC-QAR-I, Picatinny Arsenal, New Jersey 07806-5000".

PAGE 26

* 6.7: Delete "ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-Q and DRDAR-LCU-CA" and substitute "ARDEC, Picatinny Arsenal, New Jersey 07806-5000, ATTN: AMSMC-QAR-Q and AMSMC-LCU-CA".

PAGES 29-35

* Delete "APPENDIX" and "FIGURE 1" in their entirety, and substitute attached replacement pages.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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APPENDIX

A PRECRACKED CHARPY FRACTURE TOUGHNESS METHOD FOR HF-1 STEEL

10. SCOPE

10.1 Scope. This test method covers the determination of a fracture toughness, K_q, of a Charpy-type specimen (see Fig. 1) containing a fatigue crack tested in slow bending. This test method will be applicable for HF-1 steel only.

This fracture toughness will be sensitive to changes in the plane-strain fracture toughness providing the strength of the specimen is determined primarily by crack propagation and not by plastic instability in the Charpy-type specimen.

20. APPLICABLE DOCUMENTS

20.1 ASTM Standards:

E-4 Practices for Load Verification of Testing Machines
E-8 Methods of Tension Testing of Metallic Materials
E-23 Methods for Notched Bar Impact Testing of Metallic Materials
E-139 Practice for Conducting Creep, Creep-Rupture, and Stress-Rupture Tests of Metallic Materials
E-399 Test Method for Plane-Strain Fracture Toughness of Metallic Materials
E-812 Crack Strength of Slow-Bend Precracked Charpy Specimens of High-Strength Metallic Materials

30. SUMMARY OF METHOD

30.1 Summary of method. This test method employs a Charpy specimen provided with a sharp notch terminating in a fatigue crack tested in three-point bending using fixtures that minimize the contribution of friction forces to the measured applied load. The maximum load in the test is recorded and the fracture toughness is determined from this value and the original dimensions of the specimen.

40. PRECAUTIONARY NOTES

40.1 Precautionary notes. When preparing specimens and testing for fracture toughness, it is important that certain precautions be kept in mind.

40.1.1 Toughness. The fracture toughness of some materials will be sensitive to the rate of loading and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature. The fracture toughness will be influenced by the testing temperature and calibration relations may change with testing temperature.

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50. DEFINITIONS

50.1 Fracture toughness, K_q. As determined by this method, the fracture toughness is defined as the nominal stress intensity in a notched and cracked bend specimen calculated on the basis of the length of the initial uncracked ligament and the maximum load sustained in the test.

50.2 The following symbols are used in this test method:

a - crack depth (machined notch plus fatigue crack) (in.)
B - specimen thickness (in.)
W - specimen width (depth) (in.)
f(a/W) - power series of ratio (a/W) from ASTM E-812
P_{max} - maximum load applied (lbf)
S - span between support rollers (in.)
K_f - maximum stress intensity factor during fatigue (KSI $\sqrt{in.}$)
K_q - fracture toughness obtained in the test (KSI $\sqrt{in.}$)
E - Young's modulus of elasticity (KSI)
T - test temperature (^{o}F)

60. APPARATUS

60.1 Tension-testing machine. Conform to the requirements of Practices E-4.

60.2 Bend-test fixture. The procedure involves testing of the specimen in three-point bending. The best test fixture should provide support rollers that are free to rotate and move apart slightly during loading (for example, see the Test Method E-399 bend test fixture for general principles of design). The nominal span between support rollers shall be 1.57 in. (between 1.50 in. and 1.65 in.). If the diameter of the support rollers is between 0.30 in. and 0.39 in. there will be negligible change in span during deformation of the specimen.

60.3 Temperature control. The testing temperatures shall be controlled by use of an integral temperature conditioning chamber as part of the testing apparatus. Any suitable method may be used to temperature condition the specimen providing the region including the notched and cracked section can be uniformly maintained within $\pm 2^{o}F$ of the desired test temperature.

60.4 Temperature measurement. The temperature of the specimen during testing shall be measured by means of two thermocouples attached on either side of the notched and cracked section. These thermocouples may be spot welded in place; however, the spot weld locations shall avoid the crack path. At the test temperature, the difference between the temperatures indicated by these two thermocouples shall not exceed $1^{o}F$ and the average of these two temperatures shall be within $\pm 2^{o}F$ of the desired test temperature.

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60.4.1 Thermocouples. Temperatures shall be measured with calibrated thermocouples used in conjunction with potentiometers or millivoltmeters. Such measurements are subject to various errors; reference should be made to ASTM E-139 for a discussion of errors.

60.4.2 Calibration. The temperature measuring apparatus shall be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that the accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS, AND PREPARATION

70.1 Dimensions. The precracked Charpy bend specimen is shown in Fig. 2. The crack length, a, (starter notch depth + fatigue crack length) is nominally equal to one half the specimen width, W, and shall lie within the range between 0.45 and 0.55 W.

70.1.1 Crack starter. The crack starter shall consist of a V-notch having an included angle from 45 to 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 in. is usually necessary in order to facilitate the initiation of a fatigue crack at a low-stress intensity level.

70.2 Fatigue cracking. In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed:

70.2.1 Specimen condition. The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

70.2.2 Method. The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever bending or three-point bending may be used to produce fatigue cracks.

Cantilever bending facilitates completely reversed loading ($P_{min}/P_{max} = -1$) which aids in starting and propagating the crack. For practical purposes, the K calibration for three-point bending may be used to estimate the stress-intensity factors for the cantilever bending.

NOTE: The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane that is normal to the notched face of the specimen.

70.2.3 Crack. The fatigue crack shall extend at least 0.04 in. beyond the tip of the crack starter notch. During at least the last 0.02 in. of fatigue cracking the following will apply:

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(1) The ratio of the maximum-stress intensity in fatigue to Young's modulus, $K_{f(max)}/E$ shall not exceed 0.002 $in.^{1/2}$.

(2) The maximum stress intensity in fatigue shall be less than 60% of the stress intensity from the test, $K_{f(max)}$ 0.6 K_q.

(3) The stress ratio P_{min}/P_{max} will fall between -1 and 0.1.

80. TEST PROCEDURE

80.1 Number of tests. Samples shall be selected as specified in the basic specification.

80.2 Specimen measurement. All specimen dimensions shall be within the tolerances shown in Fig. 1.

80.2.1 B and W. Measure the thickness, B, to the nearest 0.001 in. at not less than two positions between the fatigue crack tip and the unnotched edge of the specimen and record the average value. Measure the depth W to the nearest 0.001 in. at two positions, one on each side of the crack starter notch and record the average value.

80.2.2 a. After fracture measure the initial notch plus fatigue crack length, a, to the nearest 0.5% at the following three positions: at the center of the crack front, a_3, and midway between the center and the intersection of the crack front with the specimen surfaces, a_2 and a_4. Use the average of these three measurements, $(a_2 + a_3 + a_4)/3$, as the crack length in the calculation of K_q (see 90.1). If the difference between any two of the crack length measurements exceeds 10% of the average, or if part of the crack front is closer to the machined notch root than 5% of the average, the specimen shall be discarded. Also, if the length of either surface trace of the crack, a_1 and a_5, is less than 80% of the average crack length, as defined above, the specimen shall be discarded. Record these five crack lengths.

80.3 Bend testing. Set up the bend-test-fixture so that the line of action of the applied load shall pass midway between the support roll centers within 0.04 in. Measure the span (distance between the support-roll centers) within 1% and record this value as S. This value should fall between 1.50 in. and 1.65 in.

80.3.1 Location. Locate the specimen with the crack tip midway between the roll centers to within 0.040 in., and square to the roll axes within 2°. Check the parallelism between the top face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied, light in this gap should extinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller shall be suitably adjusted.

80.3.2 Loading. Load the specimen at the rate between 700 lbf/minute and 3600 lbf/min. until the maximum load is reached, and record this value as P_{max}. Record the loading rate.

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NOTE: It is desirable to follow the load change carefully because specimens having low toughness will break abruptly and no slow decrease in load beyond the maximum will be observed.

90. CALCULATION AND INTERPRETATION OF RESULTS

90.1 K_q. Calculate a tentative fracture toughness value in KSI $\sqrt{in.}$ from the measured value of P_{max} and the initial dimensions of the specimen as follows:

$$K = (P_{max}\ S/1000\ B\ W^{3/2})\ f(a/W)$$

90.2 Validity. Determine the validity of this tentative fracture toughness by checking the validity of the precracking (70.2.3), the shape of the crack front (80.2.2), and the bend test conditions (60.4, 80.3). If these requirements are met, this tentative value will be considered valid and will be reported as the fracture toughness, K_q, of the HF-1 steel.

90.3 Fracture appearance. The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend specimen is given in Test Method E-399 and shall be used to characterize the fracture appearance of the precracked Charpy specimens.

100. REPORT

100.1 Report. The report shall include the following for each specimen tested:

100.1.1 All specimen dimensions (Section 80.2: B, W, a_1, a_2, a_3, a_4, a_5).

100.1.2 All test parameters (Section 80.3: S, Load Rate, P_{max}, temperature).

100.1.3 A check of all test validators:

100.1.3.1 Precracking.

A. $-1 \leq (P_{min}/P_{max}) \leq 0.1$

B. $K_{f(max)}/E > 0.002$ in.$^{1/2}$

C. $K_{f(max)} < 0.6\ K_q$

100.1.3.2 Crack length.

A. $a_{fatigue} \geq 0.04$ in. + a_{notch}

B. $.45W < a < .55W$

C. a_1 and $a_5 > 0.8\ a$

D. $a_i - a_j < 0.1\ a$; where i,j = 2, 3, 4

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100.1.3.3 Test conditions.

A. $1.5 \text{ in} < S < 1.65 \text{ in}$.

B. $700 \text{ lbf/minute} < \text{Load Rate} < 3600 \text{ lbf/min}$.

C. $T_{thermocouple} = T_{required} \pm 2^{o}F$

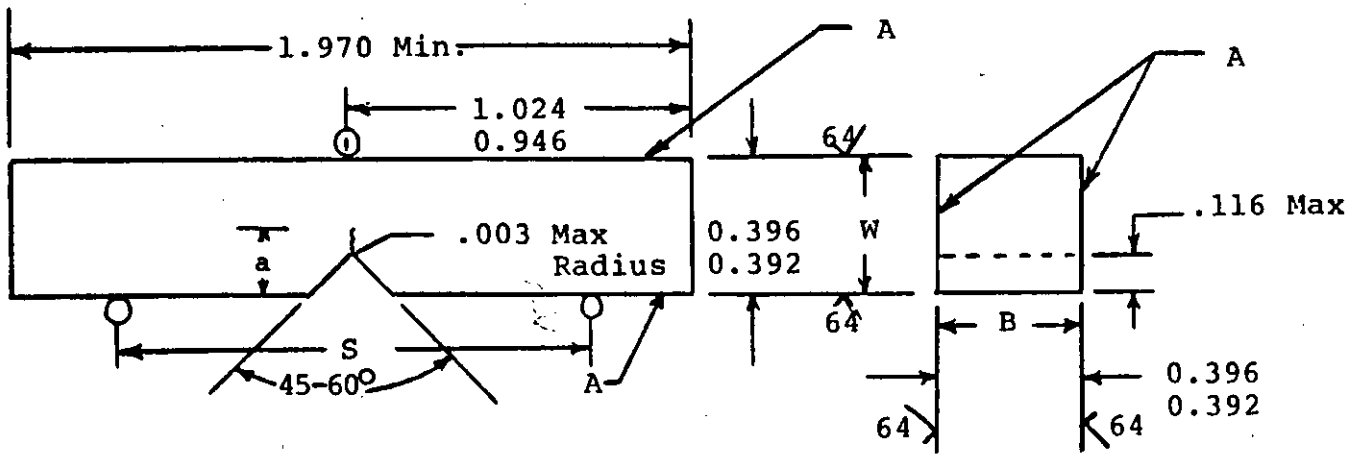
100.1.4 The fracture toughness, K_q, for all valid tests.

100.1.5 Crack plane orientation (ASTM E-399).

100.1.6 Fracture surface appearance (ASTM E-399).

100.1.7 0.2% offset tensile yield strength (ASTM E-8).

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NOTE 1. All dimensions in inches

NOTE 2. "A" surfaces shall be perpendicular or parallel as applicable to within 0.001 In.TIR

NOTE 3. Crack starter shall be perpendicular to specimen length and thickness to within ± 2 degrees.

FIGURE 2 Dimensions of precracked charpy specimens.

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